Python Lab exercise- 6

1. A college maintains a list of its students graduating every year. At the end of the year the college produces a report that lists the following.
   No of working graduates
   No of non-working graduates
   Details of the top most scorer
   Name
   Age
   Subject
   Average marks
   Write a C++ program for it that uses the following inheritance path:

   Person (name,age) → Student (rollno, average marks) → Graduate Student (subject, employed)

2. Imagine a publishing company that markets both books and audio cassette versions of its works. Create a class publication that stores the title and price of a publication. From this class derive 2 classes: book, which adds a page count and another class tape which adds playing time in minutes. Each of these 3 classes should have a getdata function() to get its data from the user at the keyboard and a putdata() function to display its data. Write a main program to test the book and tape classes by creating instances of them and calling the getdata() and putdata() functions.
3. Create a class BankAccount with account number, holder name, and balance. Include methods for deposit and withdrawal. Create a derived class SavingsAccount with an additional interest rate and a method to calculate interest. Create another derived class CurrentAccount that imposes a minimum balance. Display an appropriate message if the balance falls below the minimum.
4. Create a class Payment with a method process_payment(). Create derived classes CreditCard, UPI, and NetBanking that override the method to display different payment messages.
5. Create a base class Student with the following attributes:
   Roll number, Name, Course, Marks
   Create a derived class ScholarshipStudent that additionally stores family income.
   Write methods to:
   Calculate percentage.
   Check eligibility for scholarship.
   Display all student details.

   A student is eligible for scholarship if:
   Percentage is 80 or above, and Family income is less than ₹3,00,000.
6. Create a calculator that performs addition, subtraction, multiplication, and division.
   Handle:
   Invalid numeric input
   Division by zero
   Invalid operator

7. Write a program to accept a password. Raise an userdefined exception if the password:
   - Contains fewer than 8 characters
   - Does not contain a digit
   - Does not contain an uppercase letter

8. Accept a value from the user and convert it into an integer.

   - Use try to perform the conversion.
   - Use except to handle invalid input.
   - Use else to display the converted number.
   - Use finally to display "Program execution completed".

9. Create an exception class called InvalidAgeError. Accept the age of a person and raise this exception if the age is below 18.
10. Create a class `Student` with:

    - Roll number
    - Name
    - Marks in 5 subjects

    Write methods to:

    - Accept marks
    - Calculate total
    - Calculate percentage
    - Display grade

    Raise appropriate exceptions when:

    - Marks are outside 0–100.
    - Invalid values are entered.